

Notes: Mian and Sufi (AER, 2011)

House Prices, Home Equity–Based Borrowing, and the US Household Leverage Crisis

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1 Big picture

- **Question.** How much did rising house prices from 2002 to 2006 cause existing homeowners to borrow against home equity, and which homeowners borrowed most aggressively?
- **Setting.** The paper studies the United States housing boom and household leverage cycle before the 2007–2009 recession, using individual credit bureau data on homeowners who already owned homes as of 1997.
- **Core empirical challenge.** House price growth and borrowing may both respond to omitted shocks such as expected income growth or local productivity growth. A positive raw correlation between house prices and debt is therefore not enough to establish a causal home-equity borrowing channel.
- **Main identification idea.** Use plausibly exogenous variation in house price growth generated by differences in housing supply elasticity. When housing demand rises, inelastic housing supply areas experience larger price increases than elastic supply areas. The paper also uses a within-MSA strategy based on subprime zip-code share interacted with MSA-level supply inelasticity.
- **Main result.** The across-MSA IV estimate implies that homeowners increase borrowing by about **0.52 percent for every 1 percent increase in house prices**, or about **25 cents of borrowing for every dollar of home-value gain** from 2002 to 2006.
- **Heterogeneity.** The borrowing response is strongest among homeowners with low 1997 credit scores, high 1997 credit card utilization, and younger age. There is essentially no house-price borrowing response for homeowners in the top credit-score quartile.
- **Use of funds.** The borrowed money does not appear to be used to buy new homes, buy investment properties, or pay down credit card debt. The evidence is more consistent with consumption or home improvement.
- **Macroeconomic magnitude.** The authors estimate that home equity–based borrowing by existing homeowners explains about **\$1.25 trillion** of the rise in household debt from 2002 to 2006, about **53 percent** of the overall increase in homeowners’ debt, and at least **39 percent** of new defaults from 2006 to 2008.
- **Economic takeaway.** The housing boom mattered not only through new home buyers and risky mortgage origination. It also operated through existing homeowners, who converted rising home equity into debt, especially when they looked financially constrained or had high demand for current liquidity.

2 Data and measurement (key objects)

2.1 Observed individual-level and zip-code-level data

- **Core micro data.** Anonymous Equifax credit bureau files, observed annually from the end of 1997 through the end of 2008.
- **Initial sample.** 266,005 individuals in 3,079 zip codes located in 95 MSAs covered by Fiserv Case Shiller Weiss zip-code house price indices as of 1997.
- **Final homeowner sample.** 74,149 homeowners in 2,307 zip codes located in 68 MSAs.
- **Grouping rule.** For privacy reasons, Equifax provides the data only in groups of at least five individuals. The authors sort individuals by 1997 zip code and 1997 credit score before forming groups, so the observation is a small, relatively homogeneous group of homeowners.
- **House price data.** Zip-code-level house price indices from Fiserv Case Shiller Weiss.
- **Other zip-code data.** IRS income data, Census Business Statistics employment and payroll data, Equifax aggregate credit score data, 2000 Census demographics, and the Saiz (2010) MSA-level housing supply elasticity measure.

2.2 Homeowner definition and sample restrictions

- The credit bureau data do not directly identify homeownership. The paper classifies an individual as a 1997 homeowner if:
 - the individual has mortgage or home equity debt outstanding in 1997; or
 - the individual has a previous mortgage or home equity account on the credit report.
- This definition captures 42 percent of individuals in the credit bureau sample, lower than the census homeownership rate because the sample is at the individual level rather than household level, and because fully debt-free homeowners may not be counted.
- The authors drop homeowners who move zip codes between 1997 and 1999. This removes about 17 percent of homeowners and ensures that the assigned zip-code house price shock is meaningful before the boom.
- After these restrictions, the sample contains 74,149 homeowners in 68 MSAs.

2.3 Key dependent variables

- **Total debt growth.** Growth in all credit bureau debt between 2002 and 2006.
- **Home debt growth.** Growth in mortgage plus home equity debt.

- **Debt-to-income change.** Change in total debt divided by zip-code income.
- **Default rate.** Fraction of credit accounts in default, observed through the credit report.
- **Moving indicator.** Whether the homeowner changes zip code between 2002 and 2006.
- **Number of mortgage accounts.** Used as an indirect proxy for investment-property purchases or additional property-related borrowing.

2.4 Key explanatory variables and instruments

- **House price growth.** Zip-code or MSA house price appreciation, especially from 2002 to 2006.
- **MSA housing supply elasticity.** Saiz’s topology-based measure of how easily housing supply can expand. The measure is increasing in elasticity.
- **Housing supply inelasticity.** In the within-MSA design, the paper uses an inelasticity transformation, roughly 4 minus the Saiz elasticity measure, so higher values mean more constrained supply.
- **Subprime share.** Fraction of consumers in a zip code with credit score below 660 as of 1997. This is interacted with MSA inelasticity in the within-MSA strategy.

2.5 Descriptive facts

- Mortgage debt makes up about **88 percent** of total homeowner debt.
- Total debt is relatively flat from 1998 to 2002, then grows by about **34 percent** from 2002 to 2006.
- Home debt grows by about **39 percent** from 2002 to 2006.
- The total debt-to-income ratio rises by about **0.75** from 2002 to 2006.
- Defaults are stable from 1998 to 2006, then rise sharply from 2006 to 2008 by about **3.8 percentage points**, roughly doubling the 1997 level.

2.6 Unobservables and timing

- **Main omitted-variable concern.** Local house price growth and debt growth could both reflect local income expectations, productivity shocks, optimism, or credit-supply changes.
- **Timing.** The main boom window is 2002–2006. The later crisis/default window is 2006–2008.
- **Baseline characteristics.** Credit score, credit card utilization, debt-to-income ratio, and age are measured in 1997, before the housing boom.

- **Important design feature.** By using pre-boom borrower characteristics and supply-driven house price variation, the paper tries to isolate borrowing responses to house-price-induced collateral changes rather than reverse causality.

3 Models and empirical specifications (all equations + interpretation)

3.1 Theoretical motivation: why house prices may affect borrowing

The paper first frames the house-price borrowing response using several benchmark models.

- **Unconstrained long-lived homeowner.** If a homeowner expects to consume housing services for a long time, a higher house price raises both the value of the house and the cost of future housing consumption. In the absence of credit constraints or substitution effects, housing wealth is largely hedged, so the propensity to borrow against house-price gains should be close to zero.
- **Short-lived homeowner.** If the homeowner plans to consume part of housing wealth before death, higher house prices represent usable wealth. The borrowing response should be positive and stronger for older homeowners.
- **Credit-constrained homeowner.** If the homeowner wants to borrow but lacks collateral, a house price increase relaxes the collateral constraint. The borrowing response should be strongest for low-credit-score or high-credit-card-utilization households.
- **Self-control interpretation.** If some households have limited self-control, easier access to housing wealth may induce them to borrow for current consumption. The same observable characteristics that proxy for credit constraints may also proxy for self-control problems.

Interpretation. The paper does not need to distinguish perfectly between credit constraints and self-control. Its central empirical question is whether rising house prices causally increase homeowner borrowing and whether the effect is concentrated among financially fragile households.

3.2 Across-MSA IV specification

The main empirical specification relates homeowner leverage growth from 2002 to 2006 to instrumented house price growth:

$$LeverageGrowth_{0206,izm} = \theta X_{izm} + \beta \widehat{HP}_{0206,zm} + v_{izm}, \quad (1)$$

with first stage:

$$HPGrowth_{0206,zm} = \delta X_{izm} + \rho Elasticity_{m,1997} + \varepsilon_{izm}. \quad (2)$$

Here:

- i indexes the homeowner group;
- z indexes zip codes;
- m indexes MSAs;
- X_{izm} includes individual and zip-code controls;
- $Elasticity_{m,1997}$ is MSA-level housing supply elasticity;
- $HPGrowth$ denotes house price growth;
- β is the elasticity of borrowing with respect to house price growth.

Interpretation. The instrument uses the idea that, for a given housing demand shock, inelastic housing supply areas experience larger house price increases. If homeowners in these areas borrow more, after controls, the difference is interpreted as the causal effect of higher house prices on borrowing.

3.3 Dollar-for-dollar version

The paper also estimates a dollar version of the borrowing response:

$$\Delta Debt_{0206,izm} = \theta X_{izm} + \beta^{\$} \Delta \widehat{HV}_{0206,zm} + \eta_{izm}. \quad (3)$$

Interpretation. In this version, $\beta^{\$}$ answers the question: when home value rises by one dollar because of house price appreciation, how many cents of additional debt does the homeowner take on? The estimate is about 0.25, so the paper describes the response as roughly **25 cents of borrowing per dollar of home equity gain**.

3.4 Heterogeneity specification

To ask who borrows most aggressively, the paper interacts house price growth with pre-boom household characteristics $Z_{i,1997}$:

$$\begin{aligned} LeverageGrowth_{0206,izm} = & \theta X_{izm} + \beta \widehat{HP}_{0206,zm} \\ & + \gamma (\widehat{HP}_{0206,zm} \times Z_{i,1997}) + u_{izm}. \end{aligned} \quad (4)$$

The first stage includes both:

$$Elasticity_{m,1997} \quad \text{and} \quad Elasticity_{m,1997} \times Z_{i,1997}. \quad (5)$$

The interaction variable Z is alternately:

- 1997 credit score;
- 1997 credit card utilization;
- 1997 debt-to-income ratio;
- 2008 household income;
- 1997 age;
- male indicator.

Interpretation. This specification asks whether the borrowing response to the same house-price-induced collateral shock differs across households. The strongest patterns are for credit score, credit card utilization, and age.

3.5 Within-MSA triple-difference design

As a robustness check, the paper uses zip-code variation within MSAs. The key idea is that subprime zip codes had stronger house price growth during the boom, but this relationship appears only in inelastic MSAs.

A simplified version of the event-study/triple-difference specification is:

$$\begin{aligned} y_{izmt} - y_{izm,1998} = & \alpha_m + \delta'_t X_{izmt} + \gamma_t SubprimeShare_{zm,1997} \\ & + \beta_t (SubprimeShare_{zm,1997} \times Inelasticity_{m,1997}) + \varepsilon_{izmt}, \end{aligned} \quad (6)$$

for $t = 1999, 2000, \dots, 2008$.

Interpretation. The coefficient β_t compares high-subprime-share zip codes in inelastic MSAs to two benchmarks: high-subprime-share zip codes in elastic MSAs and lower-subprime-share zip codes in the same MSA. If debt rises in the same places and years where house prices rise, this supports the home equity borrowing channel.

3.6 Aggregate impact calculation

The paper converts the estimated elasticity into an economywide borrowing effect. Let MSAs be ordered by housing supply inelasticity, with $i = 1$ being the most elastic. Let $\widehat{\Delta P}_i$ be predicted house price growth in MSA i . The estimated house-price-induced percentage increase in borrowing is:

$$0.52 \times (\widehat{\Delta P_i} - \widehat{\Delta P_1}). \quad (7)$$

Multiplying this by initial debt and summing over homeowners gives aggregate home equity-based borrowing.

Interpretation. Because the empirical strategy is difference-in-differences-like, the paper does not identify the absolute level effect of house prices. It identifies borrowing relative to the most elastic areas, then aggregates the relative component.

4 Identification and instruments (what makes the estimates credible)

4.1 What is hard to identify

- House prices, borrowing, and local income expectations can move together.
- A positive OLS relationship could reflect expected income growth rather than collateral effects.
- Credit supply may also expand in particular places, raising both prices and borrowing.
- Existing homeowners may select into places with different long-run economic prospects.

4.2 Main instrument: housing supply elasticity

- The across-MSA design uses Saiz’s topology-based housing supply elasticity as an instrument for house price growth.
- The logic is mechanical and urban-economic: when housing demand rises, prices rise more where geography restricts new construction.
- The instrument is strong: Table 2 shows that more elastic supply predicts lower house price growth from 2002 to 2006.
- The identifying assumption is that housing supply elasticity affects borrowing only through house price growth, not through other differential shocks to borrowing demand.

4.3 Second instrument: subprime share interacted with inelasticity

- The within-MSA design uses the interaction of a zip code’s 1997 subprime share and the MSA’s housing supply inelasticity.
- The idea comes from the mortgage-credit expansion literature: credit expanded more in subprime areas, but this generated larger price increases only when housing supply was constrained.

- This allows the authors to compare zip codes within the same MSA, absorbing broad metropolitan shocks.

4.4 Evidence supporting the exclusion restriction

- The main results are robust to controls for borrower characteristics, census demographics, income growth, payroll growth, and employment growth.
- Table 5 shows little evidence that inelastic MSAs had stronger positive permanent income or employment shocks during the boom.
- The borrowing response is concentrated in home debt rather than credit card balances, consistent with borrowing against home equity.
- Renters in the same high-appreciation areas do not show comparable debt growth.
- High-credit-score and low-credit-card-utilization homeowners respond much less, which is hard to reconcile with a general local income shock story.

4.5 Remaining limitations

- The paper cannot observe direct consumption expenditures, so the use-of-funds conclusion is indirect.
- The paper cannot perfectly distinguish borrowing constraints from self-control problems.
- The main IV estimate is a local average treatment effect for house price movements generated by housing supply elasticity.
- The aggregate calculation extrapolates from the estimated effect and therefore depends on the representativeness of the credit bureau sample and the house-price data coverage.

4.6 Relation to the literature

- The paper contributes to the housing wealth and consumption literature by using individual credit bureau data and an IV strategy.
- It contributes to household finance by showing that borrowing responses are highly heterogeneous across credit quality and liquidity-use measures.
- It contributes to macro-finance by connecting asset price growth, household leverage, and subsequent default risk.
- It complements Mian and Sufi's broader work on mortgage credit expansion and the role of household leverage in the Great Recession.

5 Tables and figures

5.1 Figure 1: aggregate leverage and house price patterns

Read:

- The first panel shows that household debt rises sharply after 2002, while corporate debt rises much less.
- The second panel shows that household debt-to-income increases dramatically, whereas corporate leverage does not follow the same pattern.
- The third panel focuses on 1997 homeowners and shows that their debt rises together with aggregate house prices.
- The figure motivates the central question: was the housing boom causing existing homeowners to borrow more, or were both house prices and borrowing driven by other forces?

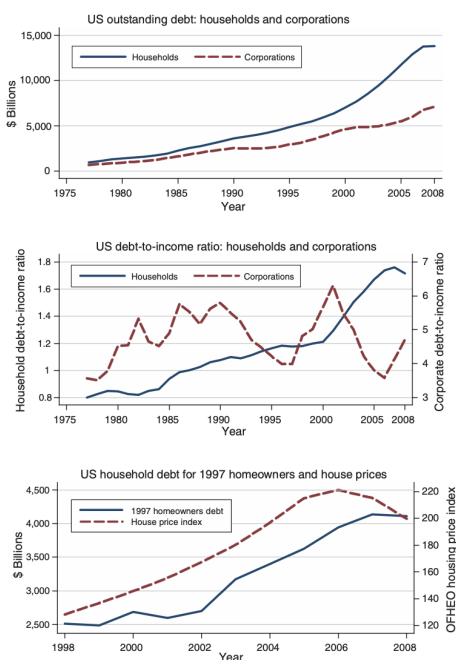


FIGURE 1. AGGREGATE US LEVERAGE AND HOUSE PRICE PATTERNS

Notes: This figure presents aggregate US leverage and house price patterns. Aggregate debt information comes from the Federal Reserve flow of funds data, aggregate income comes from National Income and Product Accounts (NIPA), and aggregate house price index data come from Office of Federal Housing Enterprise Oversight (OFHEO). In the bottom right panel, aggregate debt for 1997 homeowners comes from Equifax data where homeowners are defined to be individuals who have either an existing mortgage account with positive balance as of 1997 or a previous mortgage account.

TABLE 1—SUMMARY STATISTICS

	N	Mean	Median	SD
<i>Equifax individual-level data</i>				
Total debt, 1997 (thousands US dollars)	13,337	100	94	72
Home debt, 1997 (thousands US dollars)	13,337	88	82	70
Growth in total debt, 1998–2002	13,337	0.086	0.079	0.619
Growth in total debt, 2002–2006	13,337	0.344	0.320	0.684
Growth in home debt, 1998–2002	13,337	0.094	0.051	0.899
Growth in home debt, 2002–2006	13,337	0.394	0.355	0.901
Total debt to income ratio, 1997	13,043	2.537	2.515	1.524
Change in debt-to-income ratio, 1998–2002	13,336	−0.002	−0.085	0.903
Change in debt-to-income ratio, 2002–2006	13,336	0.754	0.424	1.381
Total debt default rate, 1997	13,337	0.038	0.000	0.111
Change in default rate, 1998–2006	13,333	−0.007	0.000	0.132
Change in default rate, 2006–2008	13,328	0.038	0.000	0.168
Credit score, 1997	13,337	780	789	95
Credit card utilization fraction, 1997	13,336	0.332	0.260	0.256
Age, 1997	13,336	48	47	7
Male	13,323	0.513	0.500	0.257
Income, 1997 (thousands US dollars)	13,336	79	73	35
<i>Fiserv Case Shiller Weiss zip-level data</i>				
House price growth, zip level, 1998–2002	13,295	0.402	0.423	0.141
House price growth, zip level, 2002–2006	13,337	0.465	0.468	0.227
<i>Saiz (2010) MSA-level elasticity measure</i>				
Housing supply elasticity	13,337	1.228	0.998	0.641
<i>Median home value, 1997, \$thousands</i>				
	13,328	129	193	135
<i>IRS zip-level income data</i>				
Per capita wage growth, 2002–2006	13,336	0.116	0.110	0.061
<i>Census business statistics zip-level data</i>				
Per capita payroll growth, 2002–2006	12,993	0.118	0.117	0.121
Employment growth, 2002–2006	12,993	0.082	0.071	0.189
<i>Equifax zip-level aggregate data</i>				
Fraction of zip code with credit score under 659, 1997	13,334	0.306	0.292	0.116

Notes: This table presents summary statistics for 74,149 individuals who have either an existing mortgage account with positive balance as of 1997 or a previous mortgage account. The sample is further restricted to individuals who do not move zip codes between 1997 and 1999. Individuals are sorted into groups of at least 5 individuals. Each group consists of individuals living in the same zip code as of 1997, and the individuals are sorted by 1997 credit score before groups are formed. The income in the denominator of the debt-to-income ratio comes from zip-level IRS data. The zip code-level median home value as of 1997 comes from the 2000 value reported in the decennial census multiplied by the growth rate from 2000 to 1997 reported in the Fiserv Case Shiller Weiss data. The housing supply elasticity measure we use is from Saiz (2010). The Saiz (2010) measure is increasing in elasticity from 0 to 12. There are 2,307 zip codes and 68 MSAs that are represented in the sample.

5.2 Table 1: summary statistics

Read:

- The analysis sample contains 74,149 homeowners, grouped into 13,337 observations.

- Median total debt in 1997 is about \$94,000, and median home debt is about \$82,000.
- Total debt growth is about 8.6 percent from 1998 to 2002 but 34.4 percent from 2002 to 2006.
- Home debt growth is even stronger, rising 39.4 percent from 2002 to 2006.
- The debt-to-income ratio rises sharply from 2002 to 2006, and defaults rise sharply after 2006.
- The table establishes that the housing boom period is also a homeowner leverage boom.

5.3 Figure 2: inelastic versus elastic housing supply MSAs

Read:

- Inelastic housing supply MSAs experience much larger house price growth from 2001 to 2006 than elastic MSAs.
- The same inelastic MSAs also experience larger total debt growth and larger increases in debt-to-income ratios.
- The figure visually summarizes the first-stage and reduced-form patterns behind the across-MSA IV strategy.
- The key comparison is not simply high-price areas versus low-price areas; it is high-price growth generated by supply constraints.

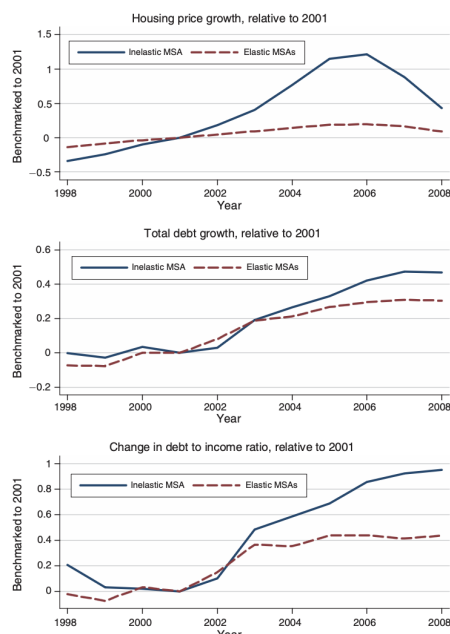


FIGURE 2. HOUSE PRICE AND LEVERAGE PATTERNS, INELASTIC VERSUS ELASTIC HOUSING SUPPLY MSAs

Notes: This figure presents house price, total debt, and total debt-to-income patterns for the top and bottom quartile MSAs based on housing supply elasticity from Saiz (2010). The sample includes individuals with mortgage debt outstanding as of 1997 or with a previous mortgage account.

Left-hand-side variable	HP growth 2002–2006 (1)	Total debt growth 2002–2006 (2)	Change in total debt-to-income ratio 2002–2006 (3)	(4)	(5)	(6)	(7)	(8)	(9)
Housing supply elasticity	–0.172*** (0.038)								
Instrumented HP growth, 2002–2006	0.519*** (0.108)	0.651*** (0.126)	0.587*** (0.118)	0.498*** (0.083)	1.473*** (0.292)	1.511*** (0.306)	1.323*** (0.270)	1.195*** (0.209)	
(Credit score, 1997)/100		–0.020* (0.010)				–0.115*** (0.028)			
Ln(household income, 2008)		0.141*** (0.019)				0.223*** (0.036)			
Debt-to-income ratio, 1997		–0.035*** (0.006)				0.065*** (0.015)			
Age, 1997		–0.013*** (0.001)				–0.030*** (0.003)			
Male dummy variable		0.031 (0.023)	0.028 (0.024)	0.018 (0.024)		0.131** (0.056)	0.125** (0.051)	0.098* (0.053)	
Additional control variables			Individual dummy variables	Individual dummy, census, income variables			Individual dummy, census, income variables	Individual dummy, census, income variables	
Observations	13,337	13,337	13,028	13,028	12,605	13,336	13,027	13,027	12,605
R ²	0.24	0.01	0.01	0.04	0.05	0.01	0.05	0.09	0.12

Notes: This table presents estimates of the effect of house prices on household borrowing for individuals who have either an existing mortgage account with positive balance as of 1997 or a previous mortgage account. Individual dummy variables are quintile indicator variables for 2 percentile bins of the 1997 credit score, 2008 income, 1997 debt-to-income ratio, and 1997 age variables. Census controls are zip code-level variables for the vacancy rate, fraction white, fraction black, education indicator variables for less than high school and high school diploma only, the unemployment rate, the poverty rate, and the fraction of households in the zip code living in an urban setting, all measured as of 2000. Income controls are zip code-level variables from the IRS and census business statistics for the logarithm of the 2002 employment, per capita wage, and per capita payroll level, and the growth in wage, payroll, and employment from 1997 to 2000, 2000 to 2002, and 2002 to 2006. All standard errors are clustered at the MSA level.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

TABLE 3—THE EFFECT OF HOUSE PRICES ON HOUSEHOLD BORROWING FOR 1997 HOMEOWNERS,
DOLLAR-FOR-DOLLAR CHANGES

Left-hand-side variable	Change in home value 2002–2006 (thousands US dollars)	Change in total debt 2002–2006 (thousands US dollars)			
	(1)	(2)	(3)	(4)	(5)
Housing supply elasticity	–50.736*** (11.179)				
Instrumented change in home value, 2002–2006		0.245*** (0.050)	0.271*** (0.056)	0.253*** (0.056)	0.246*** (0.065)
Median home value, 2002	0.459*** (0.103)	0.020 (0.039)	–0.014 (0.044)	–0.010 (0.037)	–0.076 (0.079)
(Credit score, 1997)/100			–8.176*** (1.260)		
Ln(household income, 2008)			26.960*** (3.858)		
Debt to income ratio, 1997			4.537*** (1.150)		
Age, 1997			–1.847*** (0.216)		
Male dummy variable			7.287* (4.234)	6.978* (3.996)	6.253 (4.259)
Additional control variables				Individual dummy variables	Individual dummy, census, income variables
Observations	13,328	13,328	13,199	13,199	12,497
R ²	0.6	0.03	0.07	0.1	0.12

Notes: This table presents estimates of the effect of house prices on household borrowing for individuals who have either an existing mortgage account with positive balance as of 1997 or a previous mortgage account. Individual dummy variables are quintile indicator variables for 2 percentile bins of the 1997 credit score, 2008 income, 1997 debt-to-income ratio, and 1997 age variables. Census controls are zip code-level variables for the vacancy rate, fraction white, fraction black, education indicator variables for less than high school and high school diploma only, the unemployment rate, the poverty rate, and the fraction of households in the zip code living in an urban setting, all measured as of 2000. Income controls are zip code-level variables from the IRS and census business statistics for the logarithm of the 2002 employment, per capita wage, and per capita payroll level, and the growth in wage, payroll, and employment from 1997 to 2000, 2000 to 2002, and 2002 to 2006. All standard errors are clustered at the MSA level.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

5.4 Table 2: main elasticity estimates

Read:

- Column 1 is the first stage: housing supply elasticity negatively predicts house price growth.
- Column 2 gives the main IV elasticity: a 1 percent increase in house prices raises homeowner debt by about **0.519 percent**.
- The estimate remains large and significant with individual controls, flexible individual dummy variables, census controls, and income/employment controls.
- The debt-to-income specifications also show economically meaningful effects; the estimates range from roughly 1.2 to 1.5 depending on controls.
- This table is the main causal evidence that higher house prices caused existing homeowners to increase leverage.

5.5 Table 3: dollar-for-dollar borrowing response

Read:

- The first stage shows that a decrease in housing supply elasticity strongly predicts an increase in home value.
- The IV estimates imply that a \$1 increase in home value leads to about **25 cents** of additional borrowing.

- The estimate is stable across control sets, around 0.245 to 0.271 in the main specifications.
- This table translates the elasticity estimate into a more intuitive magnitude.

5.6 Figure 3 and Table 4: heterogeneity in borrowing responses

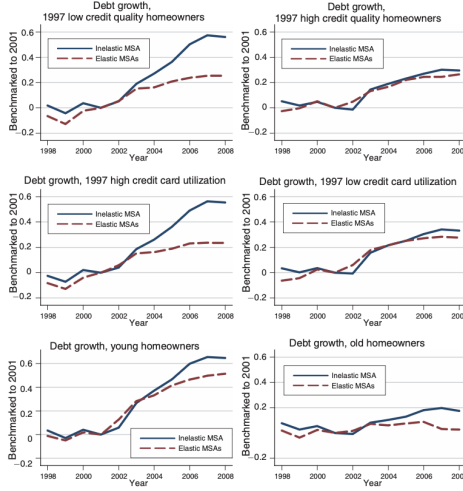


FIGURE 3. CROSS-SECTIONAL HETEROGENEITY IN LEVERAGE PATTERNS FOR 1997 HOMEOWNERS BY HOUSING SUPPLY ELASTICITY

Notes: This figure presents household leverage patterns for the highest and lowest quartile of housing supply elasticity MSAs, by the credit quality of borrowers as of 1997 (top two panels), the credit card utilization rate of borrowers as of 1997 (middle two panels), and the age of borrowers (bottom two panels). The sample includes individuals with mortgage debt outstanding as of 1997 or with a previous mortgage account.

TABLE 4—CROSS-SECTIONAL HETEROGENEITY IN EFFECT OF HOUSE PRICES ON HOUSEHOLD BORROWING FOR 1997 HOMEOWNERS

Left-hand-side variable	Total debt growth 2002–2006					
	(Credit score, 1997)/100 (1)	CC utilization, 1997 (2)	Debt to income, 1997 (3)	Ln(household income, 2008) (4)	Age, 1997 (5)	Male (6)
Instrumented house price growth, 2002–2006	2.282*** (0.497)	0.361*** (0.108)	0.356** (0.177)	0.789* (0.400)	1.416*** (0.362)	0.684*** (0.142)
Instrumented house price growth, 2002–2006 *interaction term (listed at top of column)	–0.213*** (0.059)	0.825*** (0.241)	0.130 (0.099)	–0.032 (0.085)	–0.017** (0.007)	0.016 (0.057)
(Credit score, 1997)/100	0.054* (0.032)	–0.044** (0.018)	–0.036** (0.018)	0.002 (0.031)	0.004 (0.030)	–0.055** (0.024)
Credit card utilization, 1997	–0.096* (0.056)	–0.479*** (0.106)	–0.043 (0.055)	0.048 (0.054)	0.135** (0.051)	–0.097* (0.057)
Ln(household income, 2008)	0.146*** (0.019)	0.147*** (0.019)	0.162*** (0.020)	0.202*** (0.042)	0.148*** (0.018)	0.148*** (0.021)
Debt-to-income ratio, 1997	–0.032*** (0.006)	–0.032*** (0.006)	–0.104** (0.046)	–0.044*** (0.007)	–0.041*** (0.005)	–0.037*** (0.007)
Age, 1997	–0.012*** (0.001)	–0.012*** (0.001)	–0.014*** (0.002)	–0.015*** (0.002)	–0.008*** (0.003)	–0.012*** (0.002)
Male dummy variable	0.031 (0.026)	0.031 (0.026)	0.017 (0.022)	0.003 (0.025)	–0.002 (0.029)	0.006 (0.054)
Observations	13,198	13,198	13,198	12,598	12,889	11,690
R ²	0.02	0.02	0.01	0.04	0.07	0.01

Notes: This table presents estimates of the cross-sectional heterogeneity of the effect of house prices on household borrowing for individuals who have either positive outstanding mortgage debt as of 1997 or a previous mortgage account. In each column, we interact house price growth with the variable in the top of the column. The instruments in the first stage are MSA-level housing supply elasticity and MSA-level housing supply elasticity interacted with the interaction variable listed in the top of the column. In all columns, we use the data sorts that maximize variation in the interaction variable. More specifically, in columns 1 and 2 we utilize data sorted by credit score before groups are formed. In columns 4 to 6, we utilize data sorted by each interaction variable before groups are formed. All standard errors are clustered at the MSA level.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Read:

- Figure 3 shows that the debt-growth gap between inelastic and elastic MSAs is much larger for low-credit-quality homeowners than for high-credit-quality homeowners.
- The gap is also larger for high credit card utilization homeowners than for low utilization homeowners.
- Young homeowners respond more strongly than old homeowners.
- Table 4 formalizes these patterns in IV interaction regressions.
- A one-standard-deviation higher credit score lowers the borrowing elasticity to about **0.35**; a one-standard-deviation lower credit score raises it to about **0.76**.
- A one-standard-deviation lower credit card utilization gives an elasticity around **0.36**; a one-standard-deviation higher utilization gives an elasticity around **0.78**.
- The age interaction is negative, meaning older homeowners are less responsive to house price growth. This is inconsistent with a simple life-cycle model in which older homeowners should borrow more out of housing wealth.

5.7 Table 5: examining the exclusion restriction

Read:

- Panel A checks whether housing supply elasticity is correlated with income, payroll, and employment growth. The evidence does not show a robust positive income shock in inelastic MSAs.
- Panel B shows that the effect is driven by home debt growth, not credit card balance growth.
- Renters do not show a comparable borrowing response in high-appreciation areas.
- This matters because it supports the interpretation that house prices operate through homeowners' collateral rather than through general local economic shocks.

TABLE 5—EXAMINING EXCLUSION RESTRICTION

<i>Panel A. Income and employment measures</i>						
Left-hand-side variable	Business payroll growth 2002–2006 (1)	IRS wage growth 2002–2006 (2)	Employment growth 2002–2006 (3)	Payroll growth shock (4)	Wage growth shock (5)	Employment growth shock (6)
Housing supply elasticity	–0.003 (0.006)	–0.020*** (0.003)	–0.006 (0.013)	0.016** (0.007)	–0.006 (0.005)	–0.002 (0.014)
Observations	2,341	2,306	2,341	2,220	2,306	2,220
R ²	0.00	0.04	0.00	0.00	0.00	0.00
<i>Panel B. Home, credit card, and renters' debt</i>						
Left-hand-side variable	Home debt growth 2002–2006		Credit card balances growth 2002–2006		Total debt growth 2002–2006	
	Homeowners (1)	Homeowners (2)	Homeowners (3)	Homeowners (4)	Homeowners (5)	Renters (6)
Instrumented HP growth, 2002–2006	0.535*** (0.120)	2.718*** (0.617)	0.080 (0.110)	–0.139 (0.426)	–0.029 (0.161)	0.260 (1.280)
Instrumented HP growth, 2002–2006		–0.283*** (0.073)		0.023 (0.058)		–0.041 (0.160)
*(Credit score, 1997)/100		0.093*** (0.033)		0.069*** (0.026)		–0.007 (0.081)
Observations	13,337	13,337	13,337	13,337	5,058	5,058
R ²	0.00	0.00	0.01	0.00	0.00	0.00

Notes: This table presents evidence on the exclusion restriction for the MSA-level analysis in Tables 2 and 3. Panel A examines the correlation between economic activity and housing supply elasticity in the sample of homeowners. "Growth shocks" represent the differences in 1998 to 2002 and 2002 to 2006 growth rates for each respective variable. Panel B examines the home debt and credit card balances for homeowners and debt patterns for renters. Renters are defined as individuals who have no housing related debt from 1997 to 2006. All standard errors are clustered at the MSA level.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

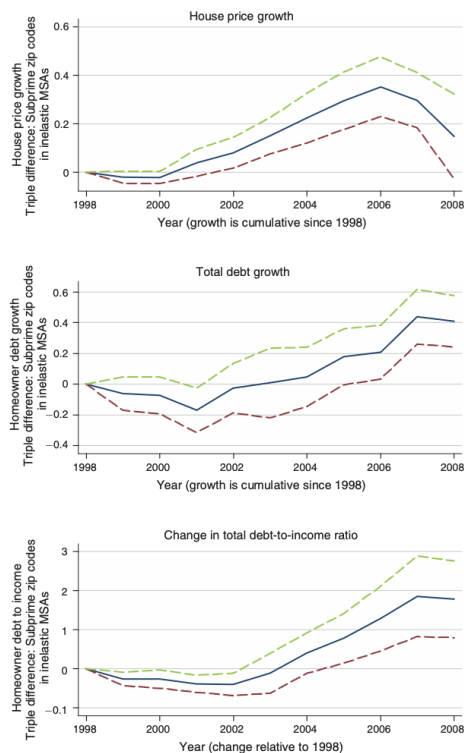


FIGURE 4. HOUSE PRICE AND LEVERAGE PATTERNS FOR HOMEOWNERS LIVING IN HIGH SUBPRIME SHARE ZIP CODES IN INELASTIC HOUSING SUPPLY MSAs

Notes: This figure plots the coefficient estimates for β^t for the following specification for each year t :

$$y_{it} - y_{it,1998} = \delta^t + \text{Subprime}_{it,1998} + \beta^t = \text{Subprime}_{it,1998} \times \text{Inelastic}_{it,1998} + \alpha_t + \varepsilon_{it},$$

where y is the natural logarithm of house prices in the top graph, the natural logarithm of total debt in the middle graph, and the debt-to-income ratio in the bottom graph. The graphs show differential patterns for homeowners living in high subprime share zip codes in inelastic housing supply MSAs relative to both homeowners living in low subprime share zip codes in the same MSA and homeowners in high subprime share zip codes in elastic housing supply MSAs.

5.8 Figure 4: within-MSA triple-difference evidence

Read:

- The top panel shows that high-subprime-share zip codes in inelastic MSAs have especially strong house price growth after 2002.
- The middle and bottom panels show that these same zip codes experience stronger debt growth and debt-to-income growth.
- The design compares high-subprime zip codes in inelastic MSAs to both high-subprime zip codes in elastic MSAs and lower-subprime zip codes in the same MSA.
- The figure is important because it shows that the main finding is not only an across-MSA pattern.

5.9 Table 6: what homeowners do with borrowed money

Read:

- Panel A shows no evidence that house price growth makes homeowners more likely to move to a new zip code.
- Debt growth responses are similar for movers and nonmovers, suggesting the effect is not mainly about trading up to a larger home.
- Panel B shows no evidence that homeowners in high-appreciation areas increase the number of mortgage accounts; if anything, the relationship is negative.
- Panel C focuses on households with high credit card utilization. Even though they borrow strongly against home equity, they do not pay down credit card balances.
- The likely use of funds is therefore consumption or home improvement, although the paper cannot observe spending directly.

5.10 Figure 5: defaults after the borrowing boom

Read:

- During the boom, default rates fall more in inelastic MSAs, especially among low-credit-quality homeowners.
- After 2006, this reverses sharply: homeowners in inelastic MSAs experience a much larger default increase.
- For low-credit-quality borrowers, defaults fall by nearly 5 percentage points from 2001 to 2005 in inelastic MSAs, but then rise by about 12 percentage points from 2006 to 2008.
- In elastic MSAs, the comparable post-2006 increase for low-credit-quality borrowers is about 4 percentage points.

TABLE 6—WHAT DO HOMEOWNERS DO WITH BORROWED MONEY?

Panel A. Purchase of new homes?				
Sample	MSA averages	Full	Total debt growth	
Left-hand-side variable	Probability of moving (1)	Probability of moving (2)	Movers (3)	Nonmovers (4)
HP growth, 2002–2006	0.046 (0.036)			
Instrumented HP growth, 2002–2006		0.010 (0.076)	0.668*** (0.140)	0.662*** (0.134)
Observations	68	13,196	6,673	6,523
R ²	0.40	0.02	0.01	0.01
Panel B. Purchase of investment properties?				
Sample	Full	Full	No change in no. of mortgages	
Left-hand-side variable	Change in no. of mortgages 2001–2005 (1)	Change in no. of mortgages 2001–2005 (2)	Full Total debt growth 2002–2006 (3)	Full Total debt growth 2002–2006 (4)
HP growth, 2002–2006	−0.011 (0.021)			
Instrumented HP growth, 2002–2006		−0.109** (0.047)	0.624*** (0.097)	0.642*** (0.138)
Change in no. of mortgages, 2001–2005			0.633*** (0.030)	
Observations	12,772	12,772	12,772	3,827
R ²	0.02	0.02	0.10	0.02
Panel C. Paying down credit card balances?				
Sample	Top quartile credit card utilization as of 1997			
Left-hand-side variable	Home debt growth 2002–2006 (1)	Credit card balance growth 2002–2006 (2)	Change in home debt to income 2002–2006 (3)	Change in credit card balance to income 2002–2006 (4)
Instrumented HP growth, 2002–2006	0.750*** (0.153)	0.084 (0.143)	1.875*** (0.394)	0.017 (0.022)
Observations	3,233	3,233	3,233	3,233
R ²	0.02	0.01	0.06	0.01

Notes: Panel A examines whether households in high house price growth areas are more likely to move to a new zip code and whether movers and nonmovers experience differential growth rates in debt from 2002 to 2006. Panel B examines whether households in high house price growth areas are more likely to increase their number of mortgages, which is a proxy for the purchase of an investment property. Panel C isolates the sample to individuals in the top quartile of the 1997 credit card utilization distribution and examines whether households in high house price appreciation areas are more likely to pay down credit card balances. All specifications include controls for 1997 credit score, 2008 income, 1997 debt to income, age, and sex. Standard errors are clustered at the MSA level.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

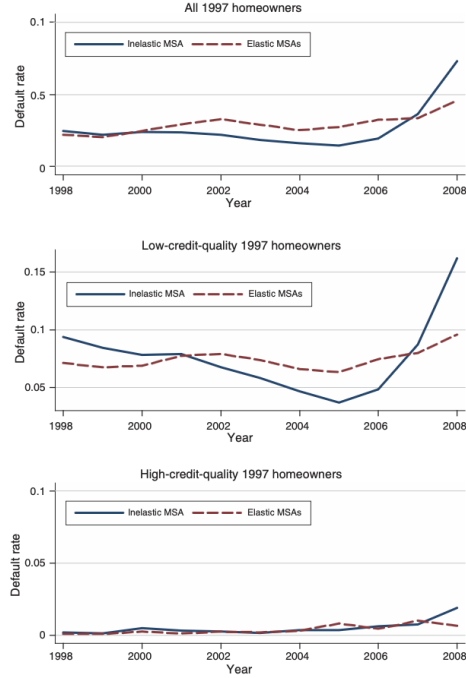


FIGURE 5. DEFAULT PATTERNS FOR HOMEOWNERS, BY HOUSING SUPPLY ELASTICITY AND 1997 CREDIT QUALITY

Notes: This figure presents default patterns for the highest and lowest quartile of housing supply elasticity MSAs, by the credit quality of borrowers as of 1997. The top graph examines the full sample. The middle graph examines borrowers in the lowest quartile of the 1997 credit score distribution, and the bottom graph examines borrowers in the highest quartile of the 1997 credit score distribution. The sample includes individuals with mortgage debt outstanding as of 1997 or with a previous mortgage account.

- The figure connects the borrowing boom to the later default crisis.

5.11 Aggregate impact calculation

Read:

- Using the main elasticity estimate of 0.52, the paper calculates that house-price-induced borrowing by existing homeowners was about **\$1.25 trillion** from 2002 to 2006.
- This accounts for about **53 percent** of the overall increase in debt among homeowners during that period.
- Using the default estimates, the authors calculate that this home equity borrowing channel accounts for at least **39 percent** of new defaults from 2006 to 2008.
- The crisis was therefore not only about marginal new home buyers entering the market; it was also about incumbent homeowners extracting equity and later defaulting.

6 Short summary

- **Method.** The paper estimates the causal effect of house price growth on homeowner borrowing using individual credit bureau data and IV strategies based on housing supply elasticity.

- **Identification.** The main instrument is MSA housing supply elasticity; a second within-MSA strategy uses 1997 subprime share interacted with supply inelasticity.
- **Main estimate.** A 1 percent increase in house prices raises homeowner borrowing by about 0.52 percent, or roughly 25 cents for every dollar of home-equity gain.
- **Heterogeneity.** Low-credit-score, high-credit-card-utilization, and younger homeowners borrow much more aggressively against rising housing wealth.
- **Use of funds and crisis link.** Borrowed funds are not used to buy real estate or pay down credit cards. The resulting leverage helps explain a substantial share of subsequent defaults.

7 Conclusion

7.1 Core conclusion

- Rising house prices from 2002 to 2006 caused existing homeowners to borrow substantially more.
- The borrowing response was highly concentrated among financially fragile or liquidity-hungry households.
- The home equity borrowing channel was large enough to matter at the macroeconomic level: it explains about \$1.25 trillion of additional debt and a significant share of later defaults.

7.2 Economic intuition

- A house price boom raises collateral values.
- For unconstrained households, higher house prices do not necessarily create spendable wealth because future housing costs also rise.
- For constrained or impatient households, higher collateral relaxes borrowing limits or increases access to liquid funds.
- The boom therefore converts asset-price appreciation into leverage, especially among low-credit-quality households.
- When house prices stop rising, the same households become vulnerable because they have higher debt and weaker balance sheets.

7.3 Takeaway

This paper gives micro-level causal evidence for a macro-financial amplification mechanism. The housing boom did not simply make homeowners wealthier on paper. It allowed many existing

homeowners to extract equity and increase leverage. That leverage expansion was concentrated among borrowers who later experienced severe default problems, helping connect the housing boom to the household leverage crisis and the Great Recession.